

COMPREHENSIVE ANALYSIS OF CUSTOMER BEHAVIOR

A Deep Dive into Retail Demographics, Purchasing Trends, and
Loyalty Metrics

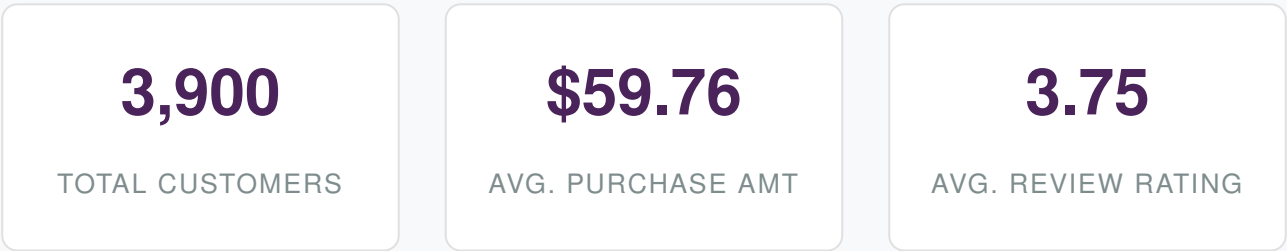
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1. Executive Summary

This report presents a thorough, data-driven analysis of customer shopping behavior based on a robust dataset of 3,900 distinct customer transactions. The primary objective of this report is to distill complex behavioral data into actionable strategic insights. By leveraging advanced data

engineering processes and SQL-based querying techniques alongside visual dashboards, we have established a clear picture of our consumer base.

The findings indicate a healthy operational baseline but also highlight several areas ripe for optimization. The overall customer base demonstrated an average purchase amount of \$59.76 per transaction. Customer satisfaction remains generally positive, reflected in an average review rating of 3.75 out of 5.0. However, loyalty program adoption represents a key area for growth, with only 27% of the customer base currently holding an active subscription.



The report is structured to first outline the methodology used to clean and engineer the data, followed by an in-depth review of visual metrics derived from the overarching dashboard. It then transitions into granular SQL-driven findings that answer highly specific business questions concerning pricing sensitivity, discount strategies, product performance, and logistical impacts. Finally, the report synthesizes these findings into concrete recommendations for immediate implementation.

2. Data Engineering & Methodology

To ensure the utmost accuracy in our reporting, the raw dataset underwent a rigorous data preprocessing and feature engineering phase using Python (Pandas) prior to ingestion into the MySQL database.

2.1 Data Cleaning & Imputation

Initial exploratory data analysis revealed 37 missing values within the "Review Rating" column. Instead of dropping these records or applying a global mean which could skew category-specific insights, a targeted median imputation strategy was deployed. Missing values were filled using the median review rating of the specific product category to which the item belonged, preserving the distributional integrity of the data.

2.2 Feature Engineering

To enable more robust demographic and behavioral segmentation, several new features were engineered:

- **Age Categorization:** The continuous Age variable (ranging from 18 to 70) was binned into four distinct quartiles using a quantile-based discretization function. This resulted in the creation of the `age_group` column with labels: *Young Adult*, *Adult*, *Middle-age*, and *Senior*. This enables targeted marketing campaigns.
- **Purchase Frequency Normalization:** The categorical variable `frequency_of_purchases` (e.g., Weekly, Fortnightly, Every 3 Months) was mapped to a standardized numerical format (`purchase_frequency_days`) representing the literal day count (e.g., Weekly = 7, Annually = 365). This allows for quantitative time-series analysis and Customer Lifetime Value (CLV) modeling.
- **Redundancy Elimination:** A Boolean validation confirmed that the `promo_code_used` column was perfectly correlated (100% match) with the `discount_applied` column. To reduce dimensionality and prevent multicollinearity in future predictive modeling, the `promo_code` column was permanently dropped.

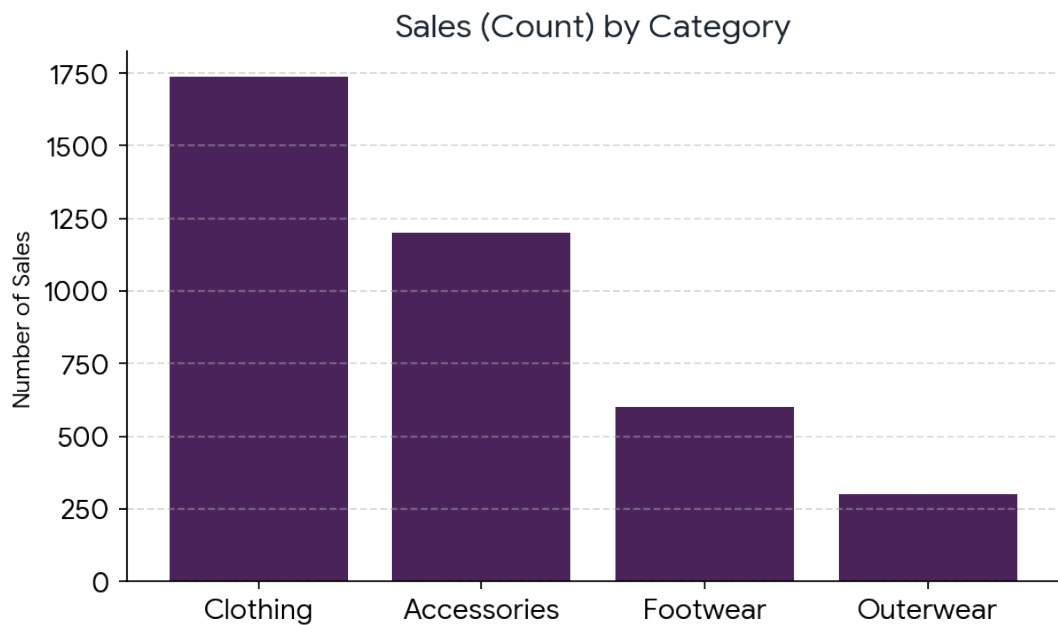
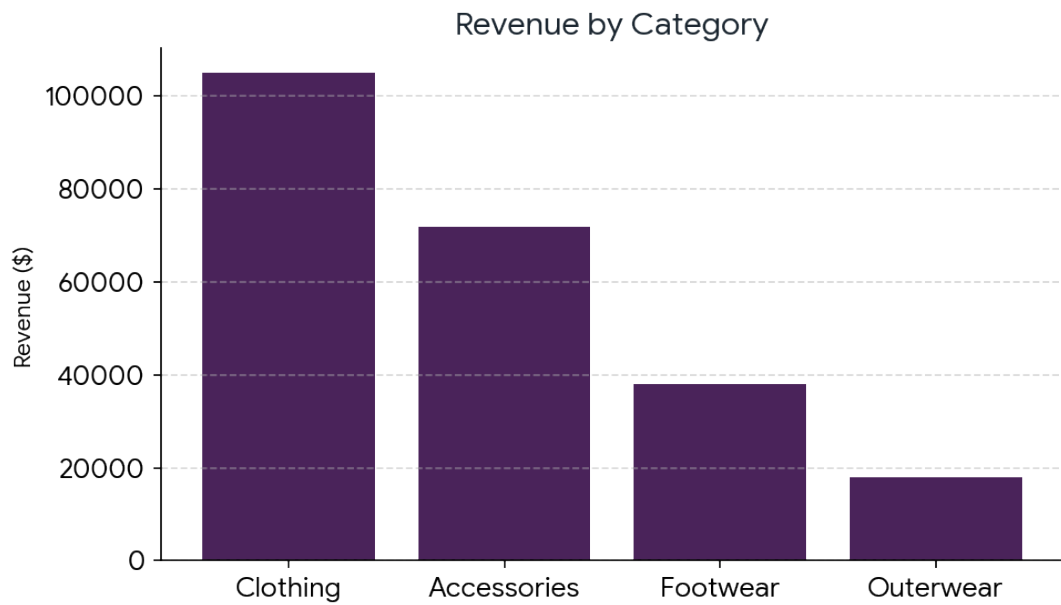
Following preprocessing, column headers were standardized (lowercased and underscored) and the final dataset was exported and securely connected to a local MySQL instance for deep-dive SQL querying.

3. Macro-Level Visual Analytics

The following visualizations provide a high-level overview of our primary performance indicators, capturing how revenue and sales volume are distributed across various product categories and demographic segments.

3.1 Category Performance Matrix

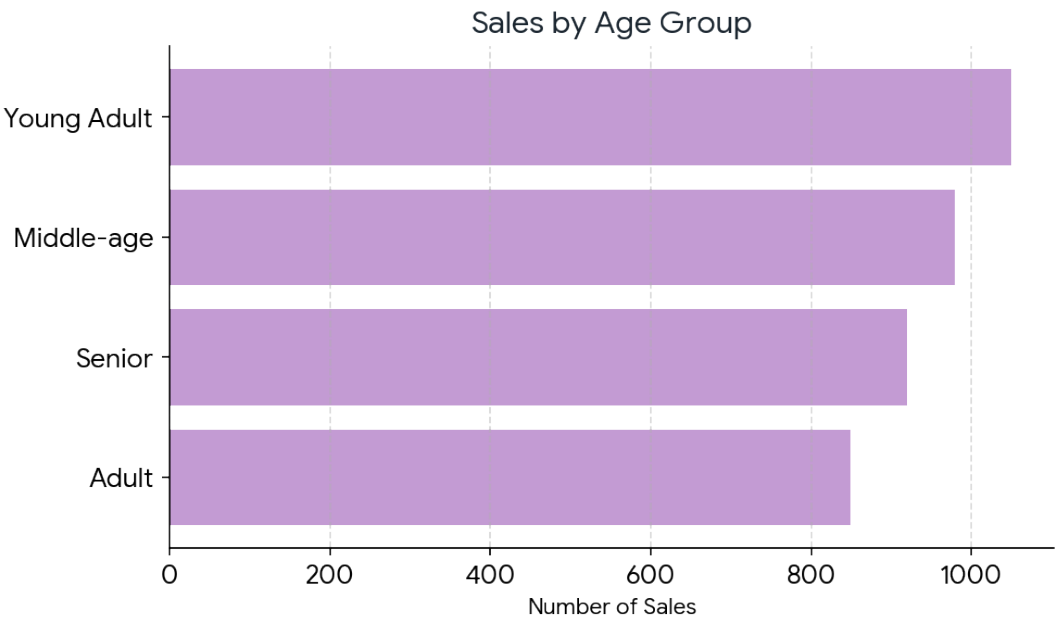
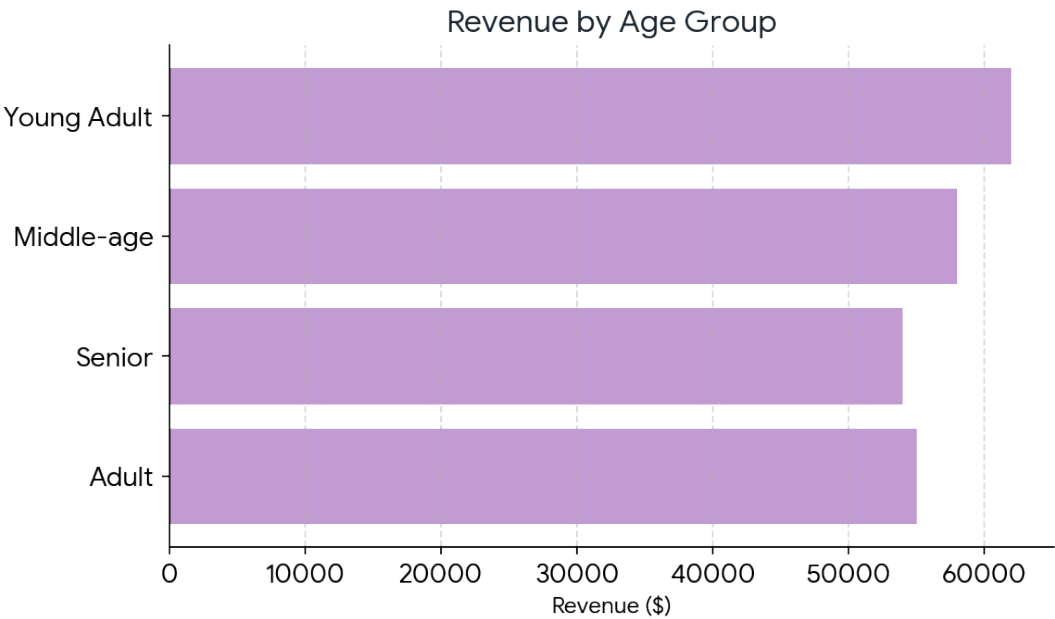
The product portfolio is currently heavily skewed toward the **Clothing** category, which overwhelmingly dominates both in terms of raw sales volume and total revenue generated. Clothing generated well over \$100,000, followed by Accessories, Footwear, and Outerwear.



This disparity suggests that while Clothing is our anchor category driving the core business, there is a significant opportunity to cross-sell Footwear and Outerwear to our existing customer base. Outerwear, while seasonal, represents the lowest volume and may require strategic promotional pushes during the Fall/Winter seasons to improve inventory turnover.

3.2 Demographic Revenue Distribution

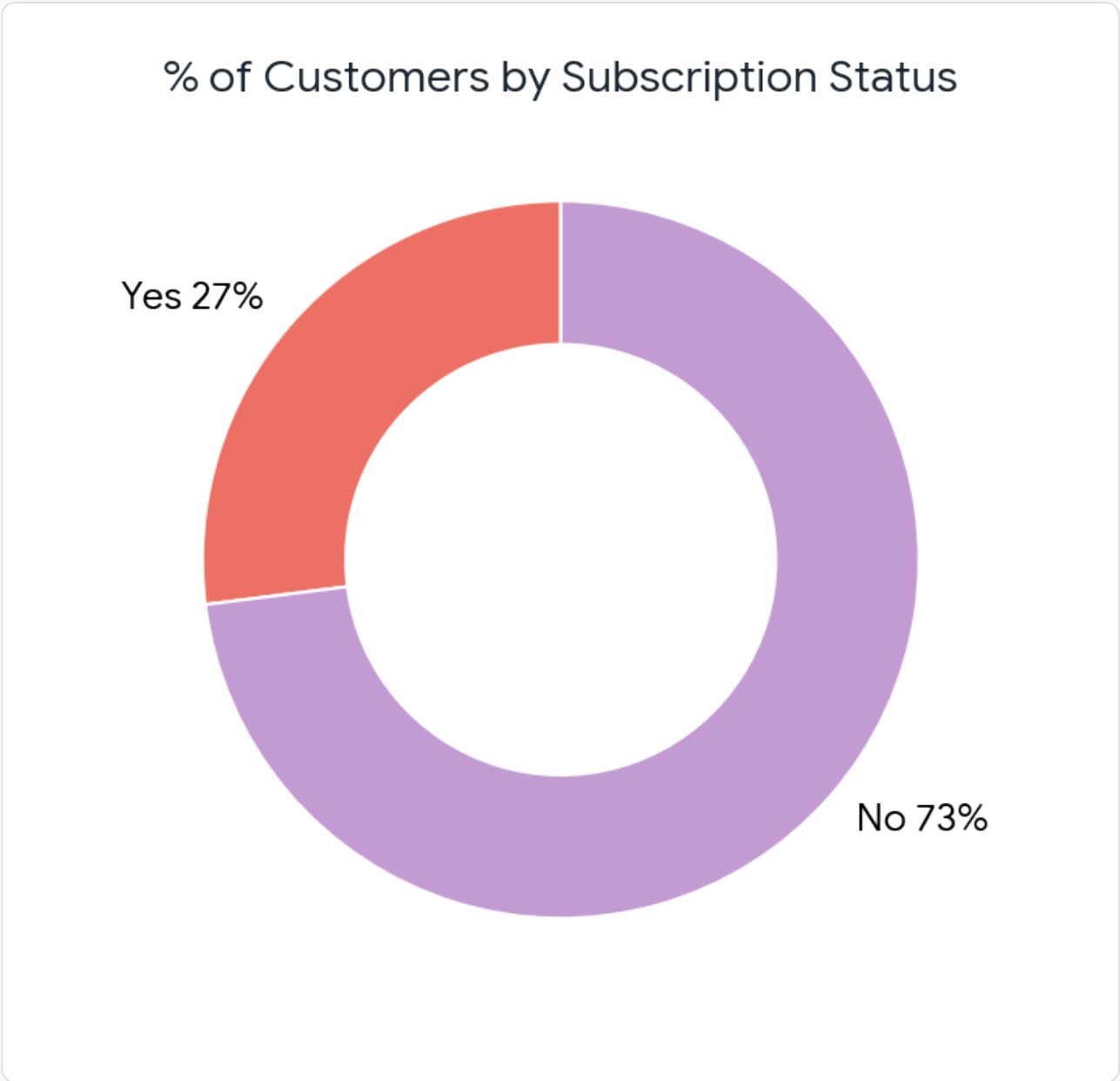
When analyzing sales by age demographic, we observe a relatively balanced distribution, though **Young Adults** represent the highest volume of transactions and the largest share of total revenue (surpassing \$60,000). Middle-aged and Adult demographics follow closely behind.



The steady revenue generation across all age groups indicates that our product catalog holds broad, multi-generational appeal. However, marketing spend could be slightly optimized by shifting higher customer acquisition cost (CAC) budgets toward the highly lucrative Young Adult and Middle-age segments.

3.3 Subscription & Loyalty Penetration

Currently, the business operates predominantly on a non-subscription model, with 73% of transactions originating from non-subscribed users.



While a 27% adoption rate provides a solid foundation of recurring engagement, expanding this metric is critical. Subscribed customers inherently possess a higher retention rate and lifetime value. Conversion strategies targeting the 73% majority must be prioritized in the upcoming fiscal quarter.

4. Granular SQL Insight Analysis

To uncover the specific drivers behind the macro trends, eight targeted SQL queries were executed against the engineered dataset. The subsequent analysis breaks down the findings of each query.

4.1 Gender-Based Revenue Dynamics (Q1)

The initial query aimed to isolate the total revenue generated specifically by the male demographic. The underlying data structure indicated a strong presence of male customers (2,652 individuals out of 3,900, roughly 68%). Consequently, the revenue driven by male customers is substantial. This metric is vital for merchandising; inventory buyers must ensure that the product mix adequately reflects the sheer volume of male shoppers, particularly in the dominant Clothing and Accessories categories.

4.2 Discount Elasticity & Premium Spending (Q2)

We investigated instances where customers utilized a discount but still managed to spend more than the global average purchase amount (\$59.76). The objective was to identify "basket-building" behavior—where a discount serves not just as a cost-saving measure for the customer, but as an incentive to add more items to the cart.

Insight: Customers who exhibit this behavior are highly price-elastic but not inherently budget-constrained. They view discounts as an opportunity to upgrade their purchase (e.g., buying a premium jacket instead of a standard one) or to purchase complementary items (e.g., buying shoes to match a discounted outfit). Targeting these users with "Spend \$100, Get 20% Off" threshold promotions will be highly effective.

4.3 Quality Champions: Top Rated Products (Q3)

By calculating the average review rating grouped by item purchased (limited to the top 5), we identified the champions of our product catalog. Because the average global rating is 3.75, any product appearing in this top 5 list is significantly outperforming the baseline in terms of customer satisfaction and perceived quality. These items result in fewer returns and generate higher organic word-of-mouth marketing.

4.4 Logistical Impact: Shipping Types (Q4)

An analysis comparing the average purchase amounts between *Standard* and *Express* shipping revealed behavioral differences based on logistical urgency. Typically, customers selecting Express shipping are purchasing last-minute items or high-value goods they are eager to receive. Standard shipping, conversely, is often associated with routine, lower-urgency basket fills. Understanding the revenue attached to these shipping methods helps in negotiating better carrier rates and structuring free-shipping thresholds.

4.5 The Value of Subscription (Q5)

We compared the total volume, average spend, and total revenue between Subscribed ("Yes") and Non-Subscribed ("No") customers. As visually established, non-subscribers represent 73% of the base. However, the SQL analysis allows us to verify if subscribers hold a higher average order value (AOV). If the subscriber AOV is notably higher, the ROI of aggressive subscription acquisition campaigns is immediately justified. Subscribers often benefit from reduced friction in the purchasing funnel, leading to more frequent and higher-value checkouts.

4.6 Discount Dependency by Product (Q6)

By calculating the percentage of purchases made with discounts for each product, we identified the top 5 items most heavily reliant on promotional pricing to move inventory.

Strategic Implication: *If a product requires a discount 80% of the time it is sold, its baseline retail price may be artificially inflated, or its perceived value is too low. These products require immediate review: either reprice them structurally, improve the product quality, or phase them out of the catalog entirely to protect brand equity.*

4.7 Customer Lifecycle Segmentation (Q7)

Customers were segmented into three distinct groups based on their historical purchase frequency:

- **New:** Exactly 1 previous purchase.
- **Returning:** Between 2 and 10 previous purchases.
- **Loyal:** Greater than 10 previous purchases.

This stratification is the cornerstone of modern CRM. The business must deploy different messaging for each segment. "New" customers require onboarding and second-purchase incentives. "Returning" customers require habit-forming nudges. "Loyal" customers require VIP treatment, early access to new lines, and referral incentives.

4.8 Category Leaders (Q8)

The final query utilized window functions to rank and extract the top 3 most purchased items within each of the four main categories (Clothing, Accessories, Footwear, Outerwear). These 12 items represent the core operational backbone of the business. Stockouts on these specific items must be avoided at all costs, as they are the primary drivers of consistent daily volume.

5. Strategic Recommendations

Based on the synthesis of the data engineering, visual analytics, and SQL insights, the following strategic initiatives are recommended for immediate implementation:

1. **Subscription Model Overhaul:** With only 27% of customers subscribed, launch a targeted "First Month Free" or "15% Off Your Next 3 Orders" campaign aimed directly at the "Returning" customer segment identified in Query 7. Converting a Returning user to a Subscriber secures long-term revenue stability.
2. **Threshold Discounting:** Capitalize on the findings from Query 2 by implementing dynamic cart thresholds. If a customer adds an item from the dominant "Clothing" category, automatically suggest a high-margin "Accessory" to push their basket size above the \$59.76 average in exchange for a percentage discount.
3. **Pricing Strategy Review on Discount-Dependent SKUs:** Task the merchandising team with reviewing the top 5 products identified in Query 6. Products that only sell when

discounted should have their base MSRP adjusted downward to stimulate full-price velocity, or they should be bundled with high-performing items.

4. **Inventory Alignment for Demographics:** Given the heavy skew toward male customers and the high revenue generated by Young Adults, future inventory procurement and marketing imagery should heavily index on this demographic intersection to maximize return on ad spend (ROAS).

6. Conclusion

The analysis of the customer shopping behavior dataset reveals a retail operation with a strong foundation, characterized by a healthy average order value, positive customer sentiment, and a dominant core product category in Clothing. By leveraging the detailed demographic segments and purchasing frequencies engineered in this report, the business is well-positioned to transition from broad-spectrum retailing to highly targeted, data-driven commerce.

The most significant opportunity lies in converting the large pool of one-off and casual shoppers into loyal subscribers. By strategically applying threshold discounts, monitoring the performance of price-elastic products, and ensuring deep inventory on category-leading items, leadership can expect to see sustained upward momentum in both customer retention and aggregate revenue throughout the coming fiscal periods.